2.3 Waves and Particle Nature of Light

This topic covers the properties of different types of wave, including standing waves. Refraction, polarisation and diffraction are also included and the wave/particle nature of light. This topic should be studied by exploring the applications of waves, for example applications in medical physics or music.

This unit includes many opportunities for developing experimental skills and techniques by carrying out more than just the core practical experiments.

Candidates will be assessed on their ability to:

33	understand the terms amplitude, frequency, period, speed and wavelength
34	be able to use the wave equation $v = f\lambda$
35	be able to describe longitudinal waves in terms of pressure variation and the displacement of molecules
36	be able to describe transverse waves
37	be able to draw and interpret graphs representing transverse and longitudinal waves including standing/stationary waves
38	**CORE PRACTICAL 4: Determine the speed of sound in air using a 2-beam oscilloscope, signal generator, speaker and microphone**
39	know and understand what is meant by *wavefront, coherence, path difference, superposition, interference* and *phase*
40	be able to use the relationship between *phase difference* and *path difference*
41	know what is meant by a *standing/stationary* wave and understand how such a wave is formed, know how to identify nodes and antinodes
42	be able to use the equation for the speed of a transverse wave on a string $v = \sqrt{\frac{T}{\mu}}$
43	**CORE PRACTICAL 5: Investigate the effects of length, tension and mass per unit length on the frequency of a vibrating string or wire**
44	be able to use the equation for the intensity of radiation $I = \frac{P}{A}$
45	know and understand that at the interface between medium 1 and medium 2 $n_1 \sin \theta_1 = n_2 \sin \theta_2$ where refractive index is $n = \frac{c}{v}$
46	be able to calculate critical angle using $\sin C = \frac{1}{n}$

47	be able to predict whether total internal reflection will occur at an interface
48	understand how to measure the refractive index of a solid material
49	understand what is meant by plane polarisation
50	understand what is meant by diffraction and use Huygens' construction to explain what happens to a wave when it meets a slit or an obstacle
51	be able to use $n\lambda = d\sin\theta$ for a diffraction grating
52	**CORE PRACTICAL 6: Determine the wavelength of light from a laser or other light source using a diffraction grating**
53	understand how diffraction experiments provide evidence for the wave nature of electrons
54	be able to use the de Broglie equation $\lambda = \frac{h}{p}$
55	understand that waves can be transmitted and reflected at an interface between media
56	understand how a pulse-echo technique can provide information about the position of an object and how the amount of information obtained may be limited by the wavelength of the radiation or by the duration of pulses
57	understand how the behaviour of electromagnetic radiation can be described in terms of a wave model and a photon model, and how these models developed over time
58	be able to use the equation $E = hf$, that relates the photon energy to the wave frequency
59	understand that the absorption of a photon can result in the emission of a photoelectron
60	understand the terms 'threshold frequency' and 'work function' and be able to use the equation $hf = \phi + \frac{1}{2}mv^2_{max}$
61	be able to use the electronvolt (eV) to express small energies
62	understand how the photoelectric effect provides evidence for the particle nature of electromagnetic radiation
63	understand atomic line spectra in terms of transitions between discrete energy levels and understand how to calculate the frequency of radiation that could be emitted or absorbed in a transition between energy levels.